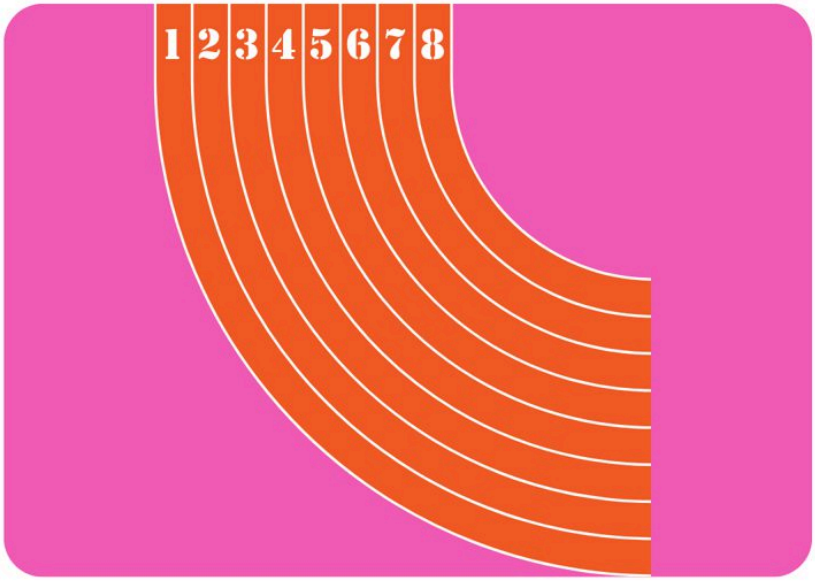
Chapter 16

Circular Motion



© James Gan

A Level Physics
Prepared and taught by James Gan

Chapter Outline

16.1 Understanding Circular Motion
16.2 Measuring Angles in Radians
16.3 Constant Speed with Changing Velocity
16.4 Angular Speed
16.5 Centripetal Forces
16.6 Determining Acceleration and Force
16.7 More on Centripetal Forces



16.1 Understanding Circular Motion

Examples of object moving in a circle



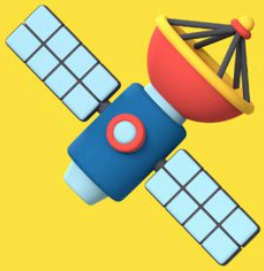
A car moving along a roundabout



The Moon orbiting the Earth



A ferris wheel at an amusement park

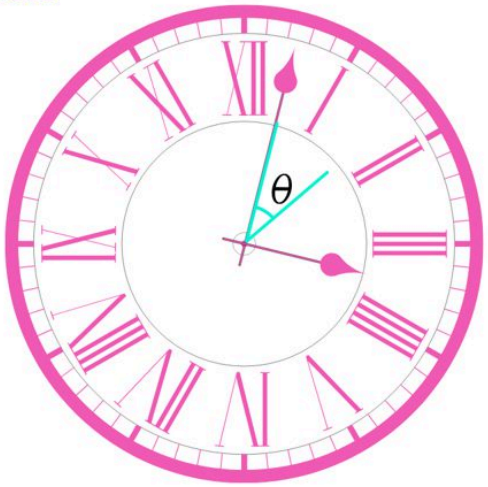


A satellite orbiting a planet

16.1 Understanding Circular Motion

Movement of the Second Hand of a Clock

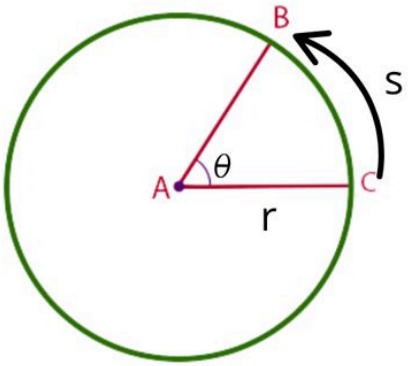
- The second hand completes 1 cycle in 60s.
- 1 cycle = 360°, so the hand moves 6°/s.
- This example shows how we can measure the motion of an object in a circle using **angles**.
- This concept leads to measuring the amount of rotation using **angular displacement**.



16.2 Measuring angles in radians

Angular Displacement

Angular displacement is the **angle** through which an object moves in a circle, measured in radians.

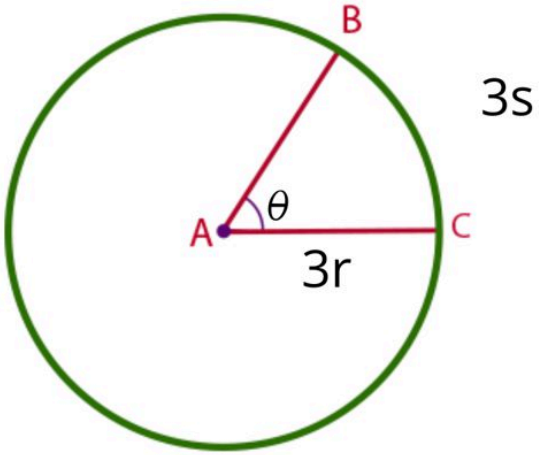


$$\theta = \frac{s}{r}$$

16.2 Measuring angles in radians

Angular Displacement

Angular displacement is the **angle** through which an object moves in a circle, measured in radians.



$$\theta = \frac{3s}{3r} = \frac{s}{r}$$

Angular displacement is simply a **ratio** of the arc length traveled to the radius of the circle.

16.2 Measuring angles in radians

Radians to Degree & v.v

Radian: A radian is the angle subtended at the center of a circle when the arc length is equal to the radius of the circle.

Formula: $\theta = \frac{s}{r}$

The angular displacement for a full circle = $2\pi r$

A complete circle contains:

$$\frac{2\pi r}{r} = 2\pi \text{ rad}$$

16.2 Measuring angles in radians

Radians to Degree & v.v

Radian: A radian is the angle subtended at the center of a circle when the arc length is equal to the radius of the circle.

Formula: $\theta = \frac{s}{r}$

The angular displacement for a full circle = $2\pi r$

A complete circle contains:

$$\frac{2\pi r}{r} = 2\pi \text{ rad}$$

2π rad = 360°

π rad = 180°

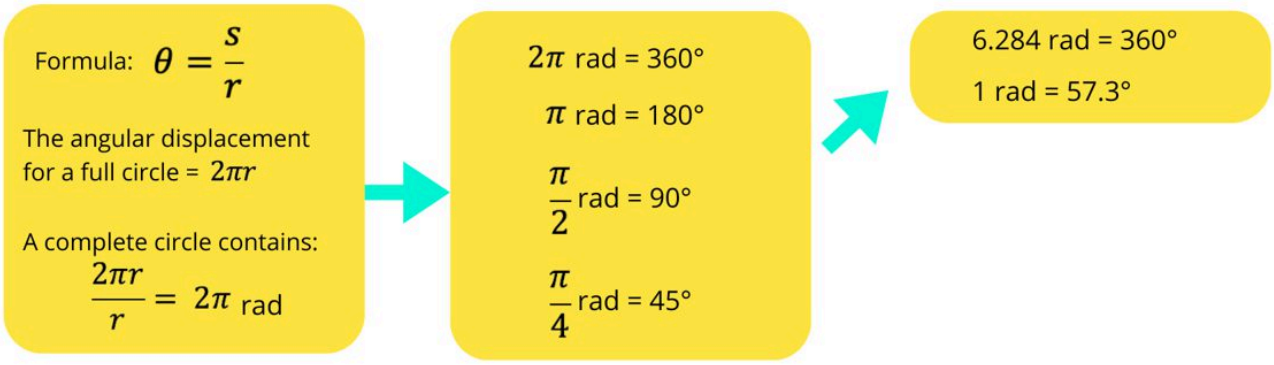
$\frac{\pi}{2}$ rad = 90°

$\frac{\pi}{4}$ rad = 45°

16.2 Measuring angles in radians

Radians to Degree & v.v

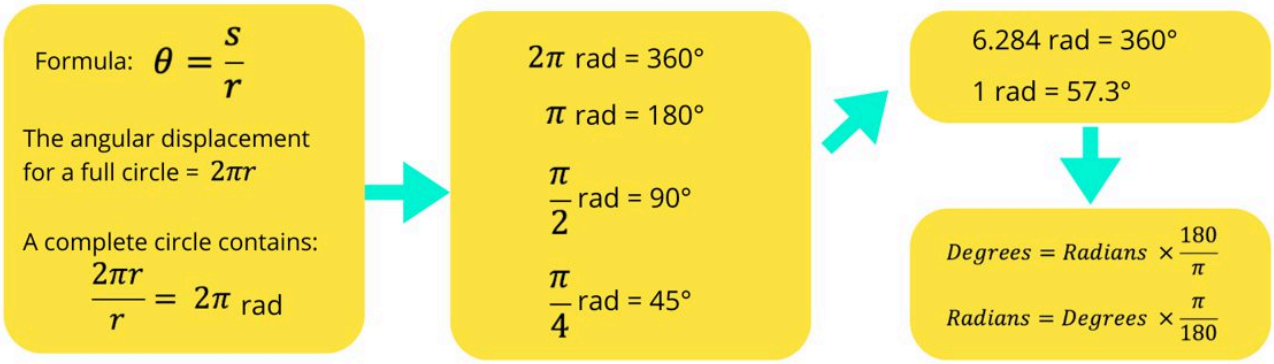
Radian: A radian is the angle subtended at the center of a circle when the arc length is equal to the radius of the circle.



16.2 Measuring angles in radians

Radians to Degree & v.v

Radian: A radian is the angle subtended at the center of a circle when the arc length is equal to the radius of the circle.



16.2 Measuring angles in radians

Worked Example

Convert 135° to radian.

16.2 Measuring angles in radians

Worked Example

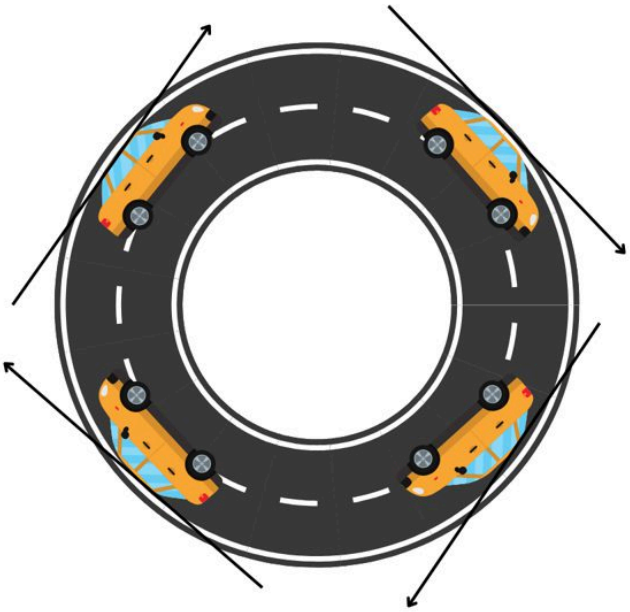
Convert 135° to radian.

$$135° \times \frac{\pi}{180} = \frac{3\pi}{4}\ radians$$

16.3 Constant Speed with Changing Velocity

Key Concept

When an object moves in a circle at constant speed, **its velocity changes** because velocity is a vector quantity that depends on both magnitude (speed) and **direction**.

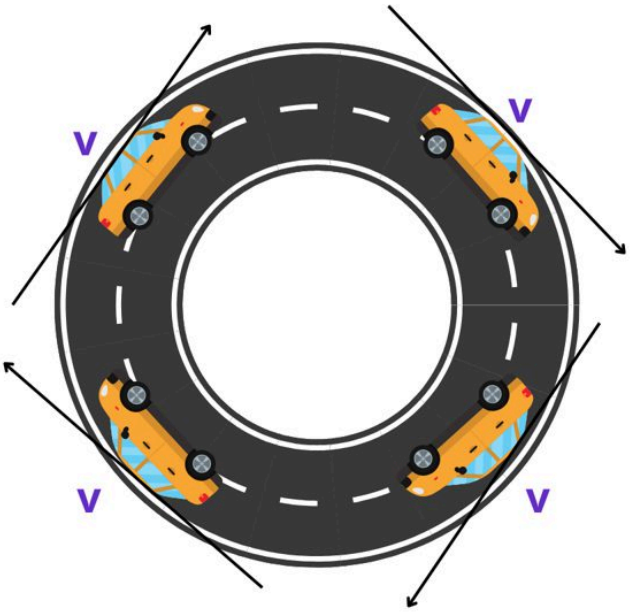


16.3 Constant Speed with Changing Velocity

Why does velocity change?

In circular motion, the direction of motion is always **changing**.

The velocity vector is always **tangential** to the circle at any given point.



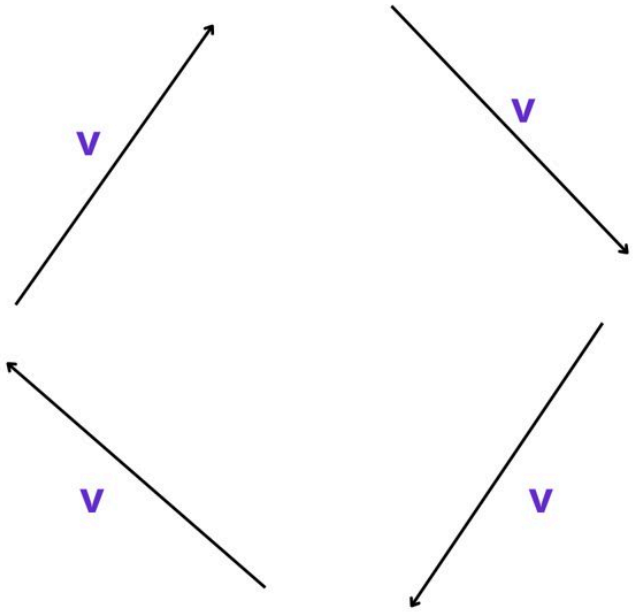
© James Gan

16.3 Constant Speed with Changing Velocity

Why does velocity change?

The tangent is a straight line that touches the circle at a single point and is perpendicular to the radius.

As the object moves along the circle, **the direction of this tangent—and thus the velocity—keeps changing.**



16.4 Angular Speed

DEFINITION

Angular speed is the rate at which an object moves through an angle in circular motion. Angular speed tells us how quickly an object rotates or revolves, regardless of its position in the circle.

FORMULA RECAP

$\theta = \frac{s}{r}$ *Angular Displacment (unit = radian)*

$s = \frac{d}{t}$ *Speed*

FORMULA

$$\omega = \frac{\Delta\theta}{\Delta t}$$

Unit: Radians per second

16.4 Angular Speed

FORMULA

$$\omega = \frac{\Delta\theta}{\Delta t}$$

Unit: Radians per second

SINGLE REVOLUTION

$$\omega = \frac{2\pi}{t}$$

16.4 Angular Speed

Worked Example

A Pirelli wheel on an F1 car completes one full revolution in 0.5 seconds.
What is its angular speed?



16.4 Angular Speed

Worked Example

A Pirelli wheel on an F1 car completes one full revolution in 0.5 seconds.
What is its angular speed?

$$\omega = \frac{2\pi}{0.5} = 4\pi \; rad/s$$



16.4 Angular Speed

Types of Speed (not in syllabus, just for clarification)

Linear speed (often simply called speed) refers to **how fast an object moves along its circular path**. It is measured in meters per second (m/s).



VS

Angular speed **(ω)** refers to **how fast an object is rotating** (in radians per second).

While angular speed tells you how much angle the object has rotated in a certain time, linear speed tells you how far the point has moved along the circular path in that same time.

16.4 Angular Speed

Relating Angular Speed and Radius to Speed

As an object moves in a circle, the linear speed of a point on the object depends on both the **angular speed (ω)** and the **radius (r)** of the circle.



KEY CONCEPT

The further a point is from the center of the circle, the faster its velocity, even if the angular speed remains constant.

16.4 Angular Speed

Relating Angular Speed and Radius to Speed

As an object moves in a circle, the linear speed of a point on the object depends on both the **angular speed (ω)** and the **radius (r)** of the circle.

FORMULA

$$v = \omega r$$

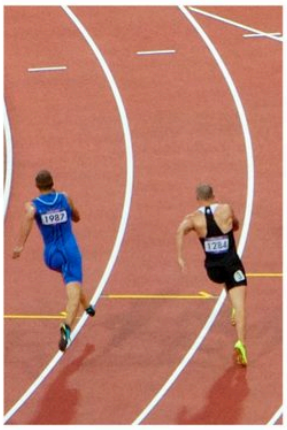
16.4 Angular Speed

Worked Example

Two runners are on a circular track. **One runner is in the inner lane, which has a radius of 50 m**, and **the other runner is in the outer lane, which has a radius of 60 m.** Both runners complete 1 full lap in 60 seconds. Calculate the linear speed for both runners.

FORMULA

$$v = \omega r$$



16.4 Angular Speed

Worked Example

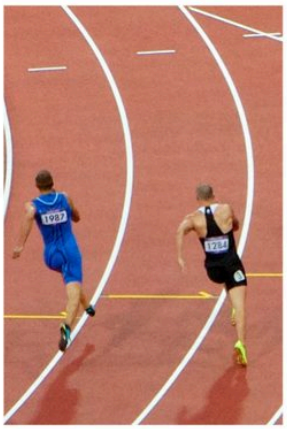
Two runners are on a circular track. **One runner is in the inner lane, which has a radius of 50 m**, and **the other runner is in the outer lane, which has a radius of 60 m.** Both runners complete 1 full lap in 60 seconds. Calculate the linear speed for both runners.

FORMULA

$$v = \omega r$$

CALCULATING ANGULAR SPEED

$$\omega = \frac{2\pi}{60} = \frac{\pi}{30}\ rad/s$$



16.4 Angular Speed

Worked Example

Two runners are on a circular track. **One runner is in the inner lane, which has a radius of 50 m**, and **the other runner is in the outer lane, which has a radius of 60 m.** Both runners complete 1 full lap in 60 seconds. Calculate the linear speed for both runners.

FORMULA

$$v = \omega r$$

CALCULATING ANGULAR SPEED

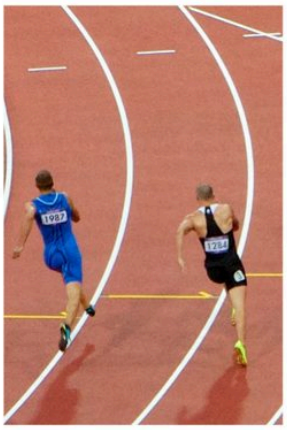
$$\omega = \frac{2\pi}{60} = \frac{\pi}{30}\ rad/s$$

INNER LANE:

$$v = 50 \times \frac{\pi}{30} = 5.24\ m/s$$

OUTER LANE:

$$v = 60 \times \frac{\pi}{30} = 6.28\ m/s$$



16.5 Centripetal Forces

Centripetal Force in Circular Motion

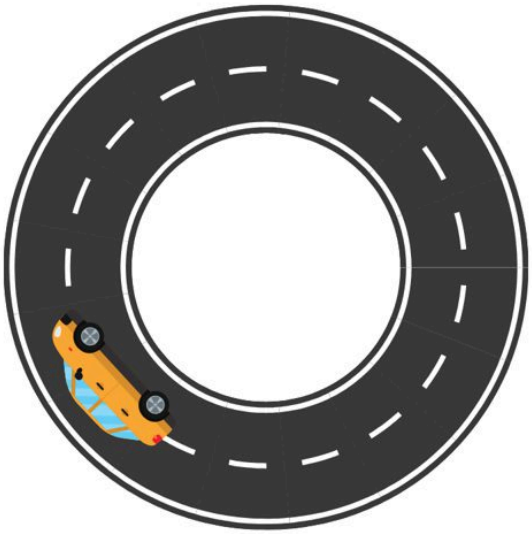
Speed is constant, but velocity changes due to direction change.

Recap

DEFINITION



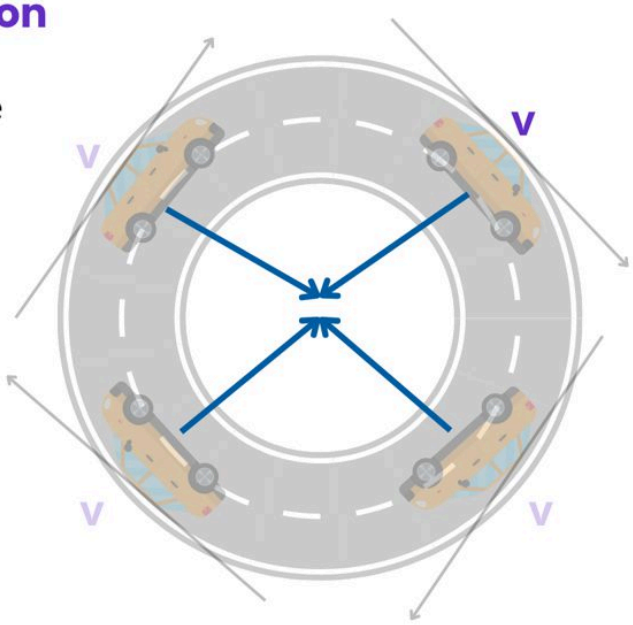
- An object at rest will **stay at rest** unless acted upon by an external force.
- An object in motion will **continue moving at a constant velocity** unless acted upon by an external force.



16.5 Centripetal Forces

Centripetal Force in Circular Motion

- This force is called centripetal force.
- It is always directed towards the center of the circle.
- Without centripetal force, the object would move in a straight line.



© James Gan

16.5 Centripetal Forces

More examples



Fidget Spinner



Ferris Wheel

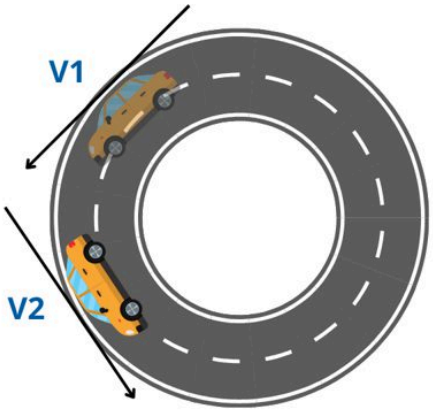
Solar System

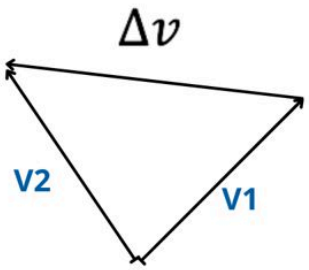


16.5 Centripetal Forces

Introduction to Acceleration in Circular Motion

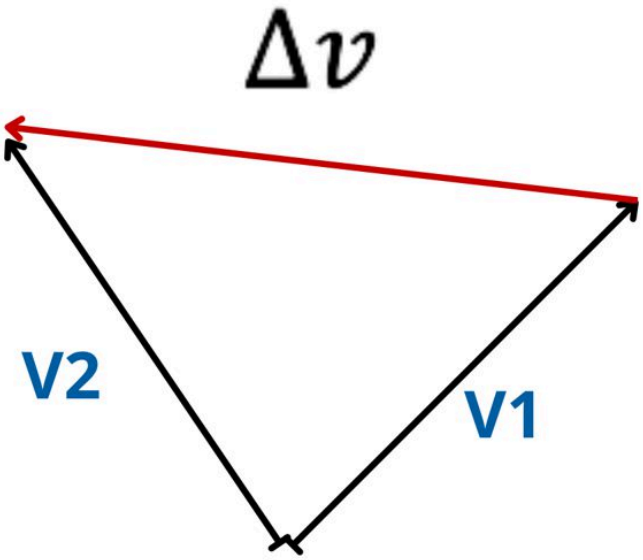
- In circular motion, velocity changes direction constantly.
- Acceleration is the rate of change of velocity.
- The change in velocity can be analyzed using a vector diagram.





16.5 Centripetal Forces

Vector Diagram and Direction of Acceleration



The **acceleration** is directed **towards the center of the circle.**

Newton's Second Law

DEFINITION



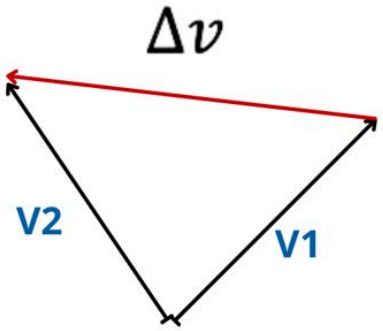
The **acceleration** of an object is
- **directly proportional** to the **net force** acting on it

Since acceleration points towards the center, **the force is also directed towards the center (centripetal force)**.

16.5 Centripetal Forces

Why Speed Remains Constant

- The speed of the object remains constant because the **force is perpendicular to the velocity.**
- The force doesn't cause any change in the object's speed, **only its direction**.
- All the force is used to **keep the object moving in a circle**, not to change its speed.



16.6 Determining Acceleration and Force

DEFINITION

Centripetal Acceleration is the acceleration that keeps an object moving in a circular path, directed towards the center of the circle.

FORMULA

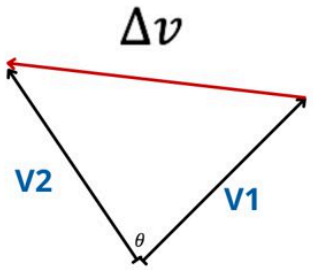
$$a = \frac{v^2}{r}$$

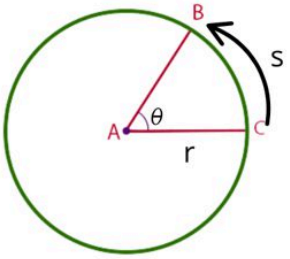
$$a = \omega^2 r$$

16.6 Determining Acceleration and Force

$v = \omega r$ $\omega = \frac{\Delta\theta}{\Delta t}$

Deriving the formula





$\theta = \frac{s}{r}$

$\Delta\theta = \frac{\Delta v}{v}$

$\frac{v\Delta\theta}{\Delta t} = \frac{\Delta v}{\Delta t}$

$\frac{v\Delta\theta}{\Delta t} = a$

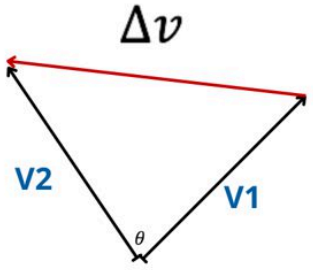
$v \cdot \frac{\Delta\theta}{\Delta t} = a$

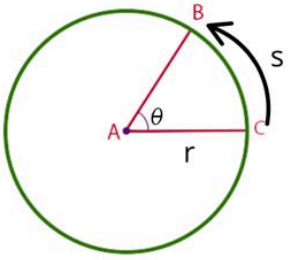
$v \cdot \omega = a$

16.6 Determining Acceleration and Force

$v = \omega r$ $\omega = \frac{\Delta\theta}{\Delta t}$

Deriving the formula





$\theta = \frac{s}{r}$

$\Delta\theta = \frac{\Delta v}{v}$

$\frac{v\Delta\theta}{\Delta t} = \frac{\Delta v}{\Delta t}$

$\frac{v\Delta\theta}{\Delta t} = a$

$v \cdot \frac{\Delta\theta}{\Delta t} = a$

$v \cdot \omega = a$

$v \cdot \left(\frac{v}{r}\right) = a$

$a = \frac{v^2}{r}$

$wr \cdot \omega = a$

$a = \omega^2 r$

16.6 Determining Acceleration and Force

Deducing a formula for centripetal force

NEWTON 2ND LAW

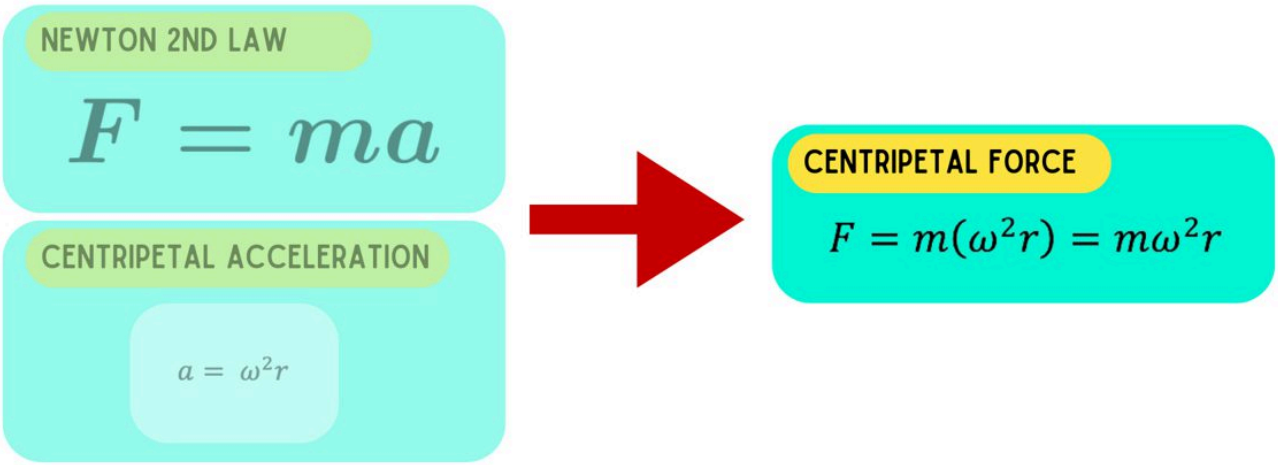
$$F = ma$$

CENTRIPETAL ACCELERATION

$$a = \omega^2 r$$

16.6 Determining Acceleration and Force

Deducing a formula for centripetal force



16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

1. Calculate the speed of the car.

16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

1. Calculate the speed of the car.

$$\omega = \frac{\Delta\theta}{\Delta t} = \frac{5 \times 2\pi}{120} = 0.262\ rad/s$$

$$v = \omega r = 0.262\ (50) = 13.1\ m/s$$

16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

2. Calculate the centripetal acceleration

© James Gan

16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

2. Calculate the centripetal acceleration

$$a = \frac{v^2}{r} = \frac{13.1^2}{50} = 3.43\ m/s^2$$

© James Gan

16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

3. Calculate the centripetal force

16.6 Determining Acceleration and Force

Worked Example

A car with a **mass of 800 kg** is moving along a circular track with a **radius of 50 m**. The car completes **5 full revolutions** in **2 minutes.**

3. Calculate the centripetal force

$$F = ma = 800\ (3.43) = 2744N$$

16.6 Determining Acceleration and Force

Finding the orbital speed

NEWTON 2ND LAW

$$F = ma$$

CENTRIPETAL ACCELERATION

$$a = \frac{v^2}{r}$$

EARTH GRAVITATIONAL PULL

$$w = mg$$

16.6 Determining Acceleration and Force

Finding the orbital speed

NEWTON 2ND LAW

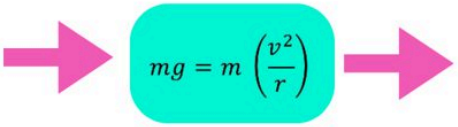
$$F = ma$$

CENTRIPETAL ACCELERATION

$$a = \frac{v^2}{r}$$

EARTH GRAVITATIONAL PULL

$$w = mg$$



$$mg = m\left(\frac{v^2}{r}\right)$$

16.6 Determining Acceleration and Force

Finding the orbital speed

NEWTON 2ND LAW

$$F = ma$$

CENTRIPETAL ACCELERATION

$$a = \frac{v^2}{r}$$

EARTH GRAVITATIONAL PULL

$$w = mg$$

$$mg = m\left(\frac{v^2}{r}\right)$$

$$mg = \frac{mv^2}{r}$$

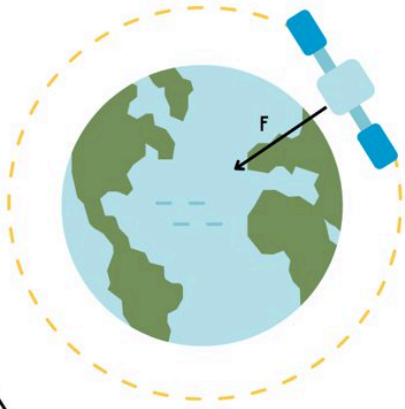
$$g = \frac{v^2}{r}$$

$$9.8 = \frac{v^2}{6400\,000}$$

$$v = 7920\ m/s$$

GRAVITATIONAL ACCELERATION

RADIUS OF THE EARTH IN METER



TO KEEP AN OBJECT IN ORBIT AROUND EARTH, IT MUST TRAVEL AT 8 KM/S. AT THIS SPEED, IT CONTINUOUSLY FALLS AROUND THE PLANET INSTEAD OF HITTING THE GROUND. AIR RESISTANCE AND GRAVITY PREVENT THROWING AN OBJECT THIS FAST, REQUIRING ROCKETS INSTEAD. 🚀

16.7 More on Centripetal Forces

Forces Acting on a Car Moving in a Circle - Vertical Force



These forces are balanced, so the car does not move vertically.

16.7 More on Centripetal Forces

Forces Acting on a Car Moving in a Circle - Horizontal Force



THE FRICTION BETWEEN THE TIRES AND THE ROAD PROVIDES THE CENTRIPETAL FORCE.

This force acts toward the center of the circular path, keeping the car in motion around the curve.

16.7 More on Centripetal Forces

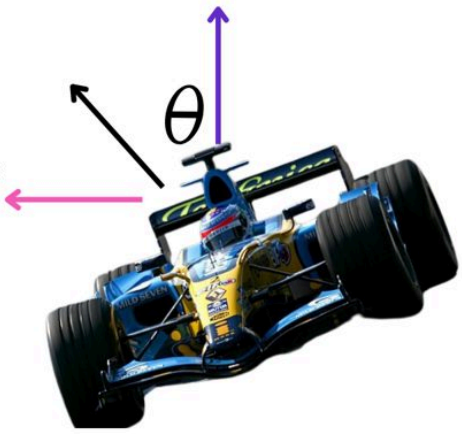
Car Cornering on a Banked Road

On a banked road, the normal force (N) is tilted.



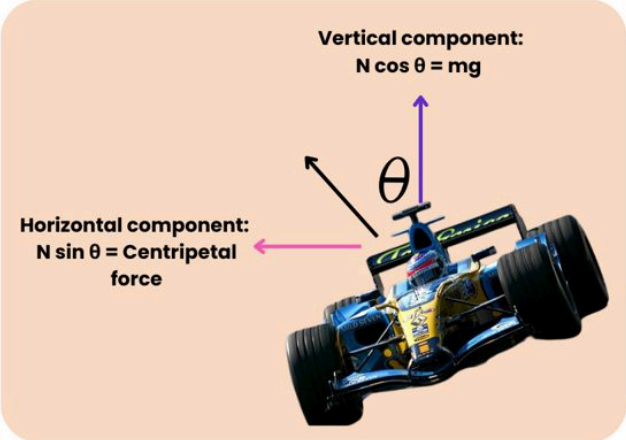
Vertical component:
N cos θ = mg

Horizontal component:
N sin θ = Centripetal force



16.7 More on Centripetal Forces

Key Takeaways



- Unlike a flat road where friction provides centripetal force, on a banked road, the tilted normal force plays a role.
- On a well-designed banking angle, **the car can turn without relying on friction**, useful for high-speed tracks.
- If the angle is too steep or the speed is too low, additional forces (friction or an engine force) may be needed.

16.7 More on Centripetal Forces

An Airplane Banking in the Sky



- When an airplane turns, it tilts its wings.
- The lift force from the wings has a **horizontal component that acts as centripetal force.**
- The vertical component still balances the plane's weight.